



National University of Computer and Emerging Sciences, Lahore Campus

Course Name:	Data Structures and Algorithms	Course Code:	CS2002
Program:	BS Electrical Engineering + BS Robotics	Semester:	Fall 2022
Duration:	1 Hour	Total Marks:	25
Exam Date:	27-Sep-2022	Weight:	15%
Section:	BEE-3A, BSR-3A	Page(s):	4
Exam Type:	Midterm-1		

- The exam is closed book and notes.
- Please answer all questions within the spaces provided.
- Use only black or blue pen to solve the paper.
- You may use page 4 for rough work. No extra sheet should be provided or collected.

Question # 1

[CLO 1, Marks: 10]

- i. [Marks: 6] Calculate the appropriate time complexity expression for the following function with respect to n and express it in terms of Big-O notation. Show all workings for full credit.

```
void mystery(int n)
{
    for (int i=1; i < n; i=i*2)
    {
        int j=0;
        while (j < 10)
            j++;
    }
}
```

Statement	frequency
$i = 1$	1
$i < n$	$\log_2 n + 1$
$i * 2$	$\log_2 n$
$\text{int } j = 0$	$\log_2 n$
$\text{while } (j < 10)$	$10 \times \log_2 n + 1$
$j++$	$10 \times \log_2 n$
$T(n) = 23 \log_2 n + 3$	
$T(n) = O(\log_2 n)$	

ii. [Marks: 4] For each function below, give an asymptotic upper bound using "big-Oh"

(a)	$f(n) = 4n^3 + 5n^2 \log n$	$O(n^2 \log n)$
(b)	$g(n) = 3n + 4 \log n$	$O(n)$
(c)	$h(n) = 1.6^n + n^5$	$O(n^5)$

List the Big-oh notations found in (i) in ascending order of complexity (fastest to slowest).

$$O(\log_2 n) > O(1)$$

(2)



Question # 2

[CLO 2, Marks: 15]

Suppose we have the following class definition of a Doubly linked list and the Node class (carrying integer data) that it composes.

<pre> class Node { private: int data; Node *next ; Node *previous ; public: Node(int pval); void setData(int pVal); int getData(); Node* getNext(); void setNext(Node* x); Node* getPrevious(); void setPrevious(Node* x); }; </pre>	<pre> class DList { private: Node *first ; public: DList(); ~DList(); //functions to be implemented //deletes the Node passed in as //parameter from the list void Delete(Node* pdel); //returns the pointer of the Node //carrying the maximum value Node* maxNode(); }; </pre>
---	--

Please implement the Delete and the maxNode member functions of the DList class in the space below taking care of all the special cases. The rest of the functions given in the above class definitions are already implemented and ready to use. Please note that since it's a doubly linked list the time complexity of the Delete function must not be worse than $O(1)$ and that of maxNode function must not be worse than $O(n)$.

void DList::Delete(Node* pdel) // 7 Marks

```

{ if ( pdel == first )
{
    first => first->getNext();
    first->setPrevious(NULL);
}
}
    
```

```

}
else
{

```

```

    Node * temp = first;
    if (Pdel != temp -> getNext())
    {
        temp = temp -> getNext();
        temp -> setNext(Pdel -> getNext());
        Pdel -> getNext(setPrevious(temp));
    }
    if (Pdel -> getNext() == Pdel -> getPrevious())
    {
        Pdel -> setNext(Pdel -> getPrevious());
    }
}
Node* DList::maxNode()// 8 Marks
{

```



```

    Node * Max = first;
    Max -> setData(Val);
    Node * temp = Max;
    Node * temp2 = Max -> getNext();

```

```

    if (Max)
    {

```

```

        while (temp > temp2)

```

```

        {

```

```

            cout << temp -> getData();

```

```

        }

```

```

        temp = temp -> getNext();

```

```

        temp2 = temp2 -> getNext();
    }
}

```

```

    return max;
}

```